SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

18CSS201J-ANALOG AND DIGITAL ELECTRONICS

QUESTION BANK
UNIT –V: REGISTERS & COUNTERS

Part – A (Each Question Carries 1 mark)

1. A register is defined as _____
   a) group of latches for storing one bit of information
   b) group of latches for storing n-bit of information
   c) group of flip-flops suitable for storing one bit of information
   d) group of flip-flops suitable for storing binary information
   Answer: d
2. The register is a type of _____
   a) Sequential circuit
   b) Combinational circuit
   c) CPU
   d) Latches
   Answer: a
3. How many types of registers are?
   a) 2
   b) 3
   c) 4
   d) 5
   Answer: c
4. Register and counter differ by the fact that _____
   a) A register has no specific sequence of states
   b) A counter has no specific sequence of states
   c) A register has capability to store one bit of information but counter has n-bit
   d) A register counts data
   Answer: a
5. For shifting to be done in one direction ,which register will be preferred _____
   a) Universal shift register
   b) Unidirectional shift register
   c) Unipolar shift register
   d) Unique shift register
   Answer: b
6. How can parallel data be taken out of a shift register simultaneously?
   a) Use the Q output of the first FF
   b) Use the Q output of the last FF
   c) Tie all of the Q outputs together
   d) Use the Q output of each FF

Answer: d

7. A 4-bit shift register that receives 4 bits of parallel data will shift to the _____ by _____ position for each clock pulse.
   a) Right, one

b) Right, two
c) Left, one
d) Left, three

Answer: a

8. To generate a sequence of equally spaced timing pulses which type of counter circuit be used ?
   a) Ring shift
   b) Clock
   c) Johnson
   d) Binary

Answer: a

9. Difference between a ring shift counter and a Johnson shift counter?
   a) There is no difference
   b) A ring is faster
   c) The feedback is reversed
   d) The Johnson is faster

Answer: c

10. How many natural states will there be in a 4-bit ripple counter?
    a) 4
    b) 8
    c) 16
    d) 32

Answer: c

11. A ripple counter's speed is limited by the propagation delay of _____
    a) Each flip-flop
    b) All flip-flops and gates
    c) The flip-flops only with gates
    d) Only circuit gates

Answer: a

12. One of the major drawbacks to the use of asynchronous counters is that _____
    a) Low-frequency applications are limited because of *internal propagation* delays
    b) High-frequency applications are limited because of internal propagation delays
    c) Asynchronous counters do not have major drawbacks and are suitable for use in high- and low-frequency counting applications
    d) Asynchronous counters do not have propagation delays, which limits their use in high-frequency applications

Answer: b

13. ……..flip-flops are required to construct a decade counter.
    a) 4
    b) 8
    c) 5
    d) 10

Answer: a

14. Which of the following statements are true?
    a) Asynchronous events does not occur at the same time

b) Asynchronous events are controlled by a clock
c) Synchronous events does not need a clock to control them
d) Only asynchronous events need a control clock

Answer: a

15. Modulus refers to _____
    a) A method used to fabricate decade counter units
    b) The modulus of elasticity, or the ability of a circuit to be stretched from one mode to another
    c) An input on a counter that is used to set the counter state, such as UP/DOWN
    d) The maximum number of states in a counter sequence

Answer: d

16. MOD-16 counter requires _____ no. of states.
    a) 8
    b) 4
    c) 16
    d) 32

Answer: c

17. High speed counter among the following is _____
    a) Ring counter
    b) Ripple counter
    c) Synchronous counter
    d) Asynchronous counter

Answer: c

18. Which of the following is a type of error associated with digital-to-analog converters (DACs)?

    a) Non monotonic error
    b) incorrect output codes
    c) offset error
    d) non monotonic and offset error

Answer: d

19. What is meant by resolution of a digital-to-analog converter (DAC)?

    a) It is the comparison between the actual output of the converter and its expected output.
    b) It is the deviation between the ideal straight-line output and the actual output of the converter.
    c) It is the smallest analog output change that can occur as a result of an increment in the digital input.
    d) It is its ability to resolve between forward and reverse steps when sequenced over its entire range.

Answer: c

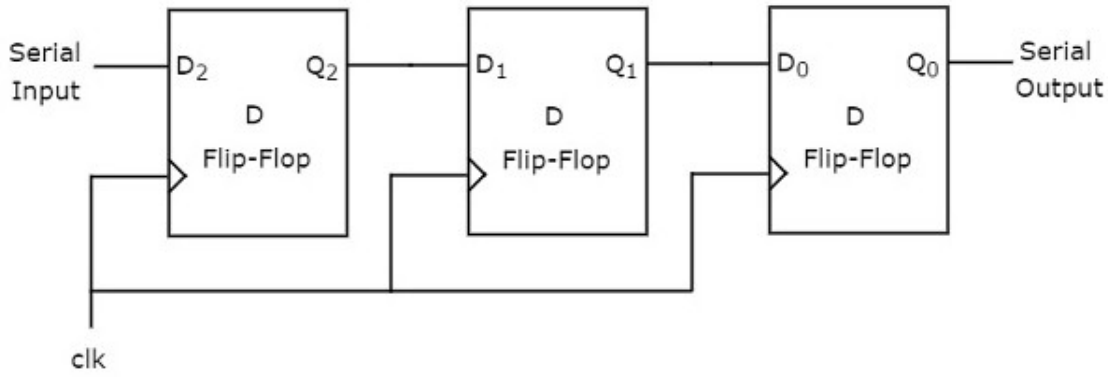
20.The practical use of binary-weighted digital-to-analog converters is limited to:

a) R/2R ladder D/A converters
b) 4-bit D/A converters
c) 8-bit D/A converters
d) op-amp comparators

Answer: b

PART – B (Each question carries 4 marks)

21. The block diagram of 3-bit SISO shift register is shown below.



How many clock pulses are required to produce produce valid output, if the binary information **"011"** from LSB to MSB serially at the input.

a) five clock pulses
b) Three clock pulses
c) four clock pulses
d) six clock pulses

Answer: a

22. Calculate time delay introduced by SISO shift register on a 4 bit digital signal, operating under frequency 4KHz .
a) 4ms
b)3ms
c)2ms
d)1ms
Answer: d

23.If the number of flip-flops used in asynchronous counter is 6,maximum number of states possible in counter is
a) 6
b)12
c)18
d)24
Answer: b

24. For a 2-bit counter to count from 00_2 to 11_2 in binary,it should have the modulus value for
a)Modulo-2
b)Modulo-4
c)Modulo-6
d)Modulo-8
Answer: b

25. In D/A converter, if no. of. bits is 6, Resolution will be
a) 0.1732
b) 2.1414
c) 1.5873
d) 0.6534
Answer: c

26. If a D/A converter is specified as the accuracy of 0.1% ,with full-scale of maximum output voltage of 10 V, the maximum error at output voltage corresponding to any input combination will be
a)1mV
b)10mV
c) 1microV
d) 10 microV
Answer: b

27. Choose the correct one from among the alternatives A,B,C,D after matching an item from Group1 with the most appropriate item in Group 2.

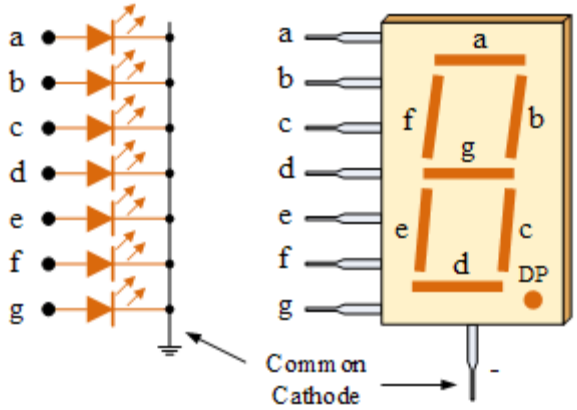
Group1	Group2
P:Shift Register	1: Frequency division
Q: Counter	2: Addressing in memory chips

R: Decoder	3: Serial to Parallel data conversion

a) P-3,Q-2,R-1
b) P-3,Q-1,R-2
c) P-2,Q-1,R-3
d) P-1,Q-2,R-2

Answer: b

28. In the common cathode display shown below, all the cathode connections of the LED segments are joined together to logic "0" or ground. Inorder to illuminate number 3,through which terminal LOGIC 1 be applied.



a)a,b,c,d,g
b)a,b,c,d
c)a,b,c,g
d)a,b,c,d,e,f

Answer: a

29. A temperature in the range of −40° C to 55° C is to be measured with a resolution of 0.1° C. The minimum number of ADC bits required to get a matching dynamic range of the temperature sensor is

a)8
b)10
c)12
d)14

Answer: b

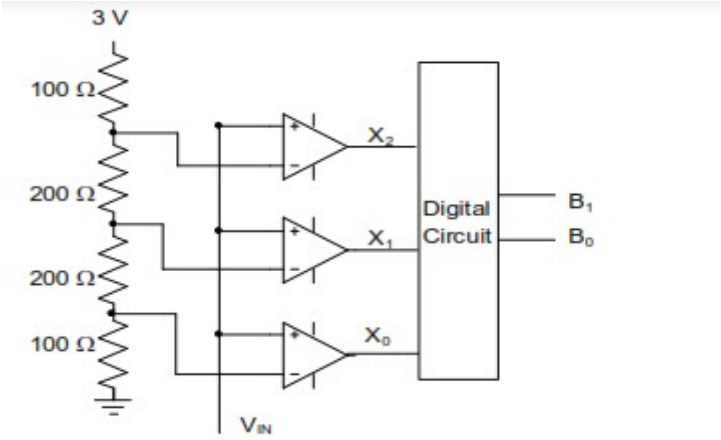
30. An 8 bit unipolar Successive Approximation Register type ADC is used to convert 3.5 V to digital equivalent output. The reference voltage is +5 V. The output of ADC at end of 3rd clock pulse after the start of conversion is _____.

a) 1010 0000
b) 1000 0000
c) 0000 0001
d) 0000 0011

Answer: a

PART – C (Each Question Carries 12 Marks)

31. A 2-bit flash Analog to Digital Converter (ADC) is given below. The input is $0 \leq V_{IN} \leq 3$ Volts.The expression for the LSB of the output B_0 as a Boolean function of X_2, X_1, and X_0 is



a) $X_0 \left[\overline{X_2 \oplus X_1}\right]$

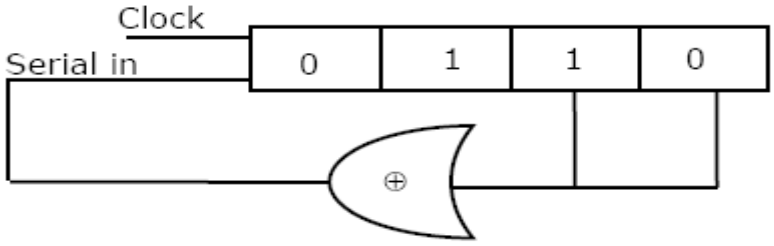
b) $\overline{X}_0 \left[\overline{X_2 \oplus X_1}\right]$

c) $X_0 [X_2 \oplus X_1]$

d) $\overline{X}_0 [X_2 \oplus X_1]$

Answer: a

32.The initial contents of the 4bit serial in serial out,right shift,shift register shown in figure,are 0110.After three clock pulses are applied, the contents of the shift register will be

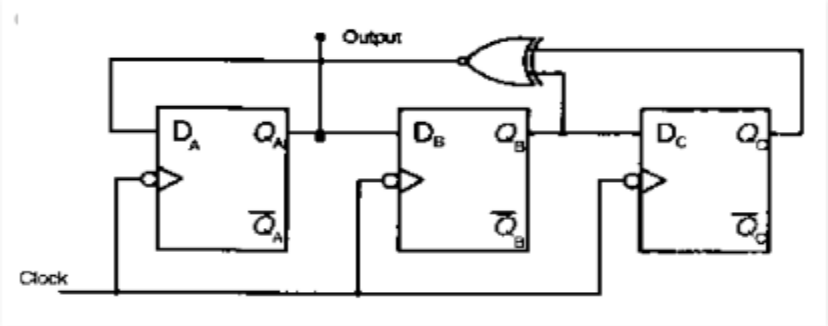


a) 0000
b) 0101

c) 1010
d) 1111

Answer: C

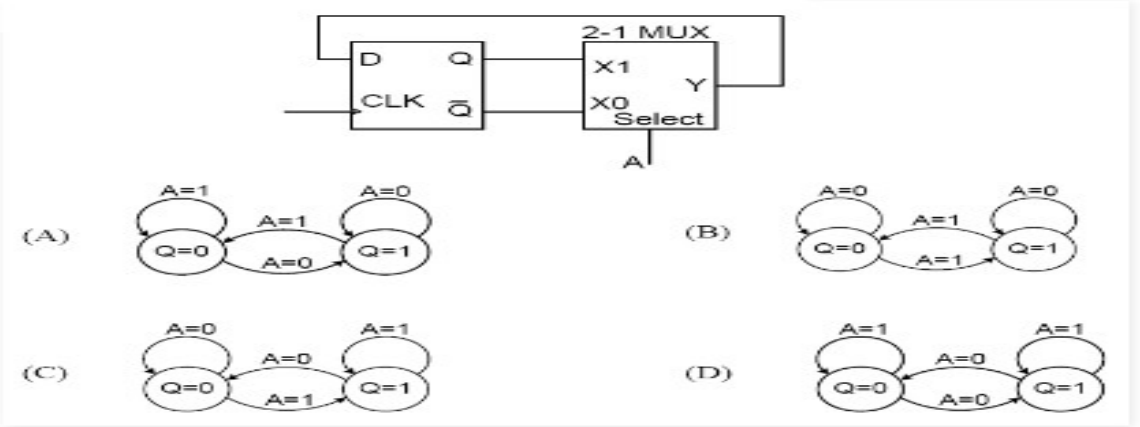
33. Assuming that all flip-flops are in reset condition initially,the count sequence observed at Q_A in the circuit shown is



a) 0010111….
b) 0001011….
c) 0101111….
d) 0110100….

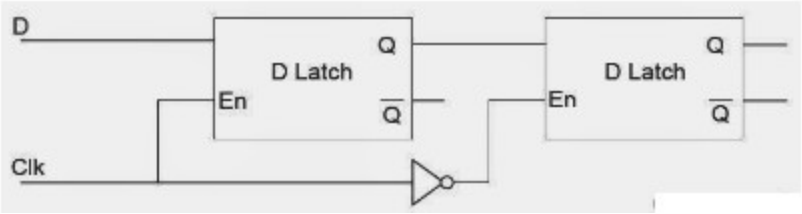
Answer: d

34. The state transition diagram for the logic circuit shown is



Answer:D

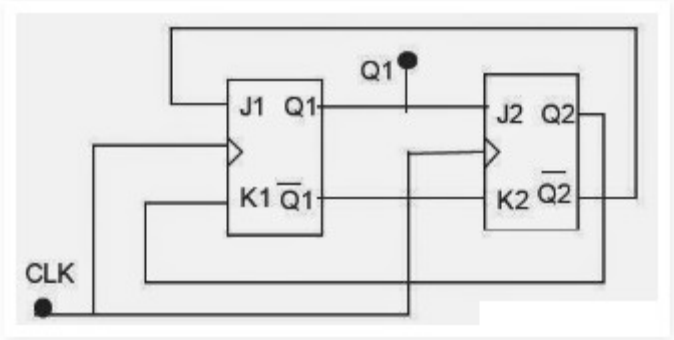
35. The circuit shown in the figure is a



a. Toggle flip flop
b. JK flip flop
c. SR latch
d. Master-slave D flip flop

Answer: d

36. The outputs of the two flip-flops Q_1 ,Q_2 in the figure shown are initialized to 0,0. The sequence generated at Q_1 upon application of clock signal is



a) 01110….
b) 01010….
c) 00110….
d) 01100….

Answer: d